eryar
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AutoDraft- Plant drawing system for AVEVA E3D and Intergraph S3D.
PlantAssistant - Translate AVEVA RVM/SP3D VUE to glTF, STEP etc.

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The intersection of Boolean data surfaces
Posted on 2023-09-29 18:45 eryar Reading(1188) Comments(0) Edit Collection Citation Classification: 2.OpenCASCADE .

The intersection of Boolean data surfaces

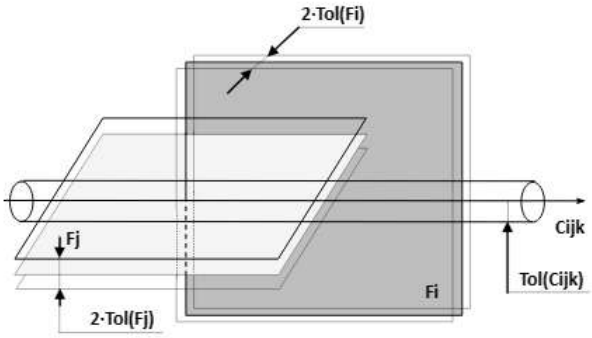
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1 Introduction

The intersection definition of opposite faces in OpenCASCADE is shown in the following image:

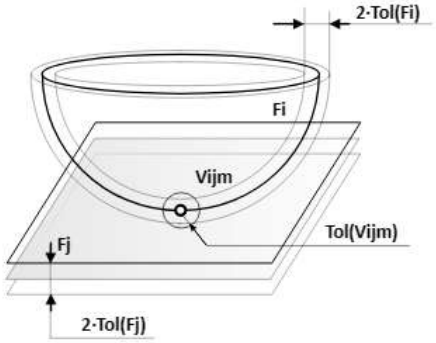
Face/Face Interference

For a face *Fi* and a face *Fj* (with the corresponding surfaces *Si* and *Sj*) there are some places in 3D space, where the distance between the surfaces is less than (or equal to) sum of tolerances of the faces.



Face/face interference: common curves

Two faces with Geometry Surfaces in three-dimensional space are considered intersecting when the distance between the two surfaces is less than the tolerance tolerance in the face. Generally, the intersection between two faces is the intersection line, and in some cases, the intersection point is obtained, as shown in the figure below:



Face/face interference: common points

The intersection of surfaces in Boolean operations is a relatively complex problem, in addition to considering the above-mentioned problems of intersection lines and intersections, it is also necessary to consider overlapping situations; For newly generated intersections, consider generating PCurve; if there is a hole on the surface, the intersection line through the opening area should be excluded; Finally, consider how to save the results of the intersection of faces. The intersection is calculated in the function:

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```
//================================================================
void BOPAlgo_PaveFiller::PerformFF(const Message_ProgressRange& theRange)
{
  // Update face info for all Face/Face intersection pairs
  // and also for the rest of the faces with FaceInfo already initialized,
  // i.e. anyhow touched faces.
  myIterator->Initialize(TopAbs_FACE, TopAbs_FACE);
  Standard_Integer iSize = myIterator->ExpectedLength();

  // Collect faces from intersection pairs
  TColStd_MapOfInteger aMIFence;
  Standard_Integer nF1, nF2;
  for (; myIterator->More(); myIterator->Next())
  {
    myIterator->Value(nF1, nF2);
    aMIFence.Add (nF1);
    aMIFence.Add (nF2);
  }
```

Ultimately, it is the IntTools_FaceFace that is called to calculate the intersection of the two faces. And save the information such as the intersection of the calculated lines, intersections, and whether they overlap in the BOPDS_InterfFF:

BOPAlgo_PaveFiller_7.cxx | BOPAlgo_PaveFiller_6.cxx | BOPAlgo_PaveFiller_1.cxx | BOPTools_AlgoTools.hxx

TKBO → BOPAlgo_PaveFiller

```
489       //
490       BOPDS_InterfFF& aFF = aFFs.Appended();
491       aFF.SetIndices(nF1, nF2);
492       aFF.SetTangentFaces(bTangentFaces);
493       aFF.Init(aNbCurves, aNbPoints);
494       //
495       // Curves
496       // Fix bounding box expanding coefficient.
497       Standard_Real aBoxExpandValue = aTolFF;
498       if (aNbCurves > 0) { ... }
506       //
507       BOPDS_VectorOfCurve& aVNC = aFF.ChangeCurves();
508       for (Standard_Integer i = 1; i <= aNbCurves; ++i) { ... }
525       //
526       // Points
527       BOPDS_VectorOfPoint& aVNP = aFF.ChangePoints();
528       for (Standard_Integer i = 1; i <= aNbPoints; ++i) {
529         const IntTools_PntOn2Faces& aPi = aPntsX(i);
530         const gp_Pnt& aP = aPi.P1().Pnt();
531         //
532         BOPDS_Point& aNP = aVNP.Appended();
533         aNP.SetPnt(aP);
534       }
```

2 Face Info

The class BOPDS_FaceInfo is used to store the following information:

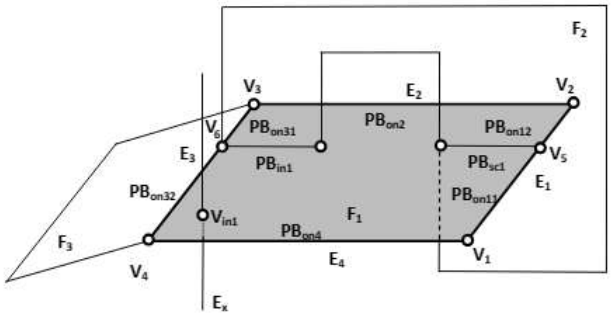
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Links

FaceInfo

The structure *FaceInfo* contains the following information:

- Pave blocks that have state **In** for the face;
- Vertices that have state **In** for the face;
- Pave blocks that have state **On** for the face;
- Vertices that have state **On** for the face;
- Pave blocks built up from intersection curves for the face;
- Vertices built up from intersection points for the face.



Face Info

Note that PBo31 and PBSc1 have one state of On and the other state of Section. The state of these intersections is updated by the function BOPAlgo_PaveFiller::UpdateFaceInfo in the intersection processing class BOPAlgo_PaveFiller.

3 Tangent Face

Judging from the imprecise logic of the previous article on using fixed sampling points to detect whether there is overlap between edges and edges and between surfaces, judging the overlap of lines is a complex problem, and judging the overlap between surfaces is relatively more complicated. This article starts with simplicity, first looks at the simplest detection of overlap between two planes, and leads to everyone's thinking about the detection of overlapping areas of any two faces. This particular case is handled in the function PerformPlanes() in IntTools_FaceFace. The geometric intersection method of quadratic surfaces is used for processing, and the source code is as follows:

```
319  void IntAna_QuadQuadGeo::Perform (const gp_Pln& P1,
320                                    const gp_Pln& P2,
321                                    const Standard_Real TolAng,
322                                    const Standard_Real Tol)
323  {
324    Standard_Real A1, B1, C1, D1, A2, B2, C2, D2, dist1, dist2, aMVD;
325    //
326    done=Standard_False;
327    param2bis=0.;
328    //
329    P1.Coefficients(A1,B1,C1,D1);
330    P2.Coefficients(A2,B2,C2,D2);
331    //
332    gp_Vec aVN1(A1,B1,C1);
333    gp_Vec aVN2(A2,B2,C2);
334    gp_Vec vd(aVN1.Crossed(aVN2));
335    //
336    const gp_Pnt& aLocP1=P1.Location();
337    const gp_Pnt& aLocP2=P2.Location();
338    //
339    dist1=A2*aLocP1.X() + B2*aLocP1.Y() + C2*aLocP1.Z() + D2;
340    dist2=A1*aLocP2.X() + B1*aLocP2.Y() + C1*aLocP2.Z() + D1;
341    //
342    aMVD=vd.Magnitude();
343    if(aMVD <=TolAng) {
344      // normalles are collinear - planes are same or parallel
345      typeres = (Abs(dist1) <= Tol && Abs(dist2) <= Tol) ? IntAna_Same
346        : IntAna_Empty;
347    }
```

From the source code, it can be seen that if the normal angle between the two planes is less than TolAng and the distance is less than Tol, the two planes are considered to be the same IntAna_Same. When the distance is greater than the Tol, it is considered that there is no intersecting IntAna_Empty.

1.OpenCASCADE Official
2.OpenCASCADE Digest
3.PipeCAD
4.Qt
5.Github/eryar

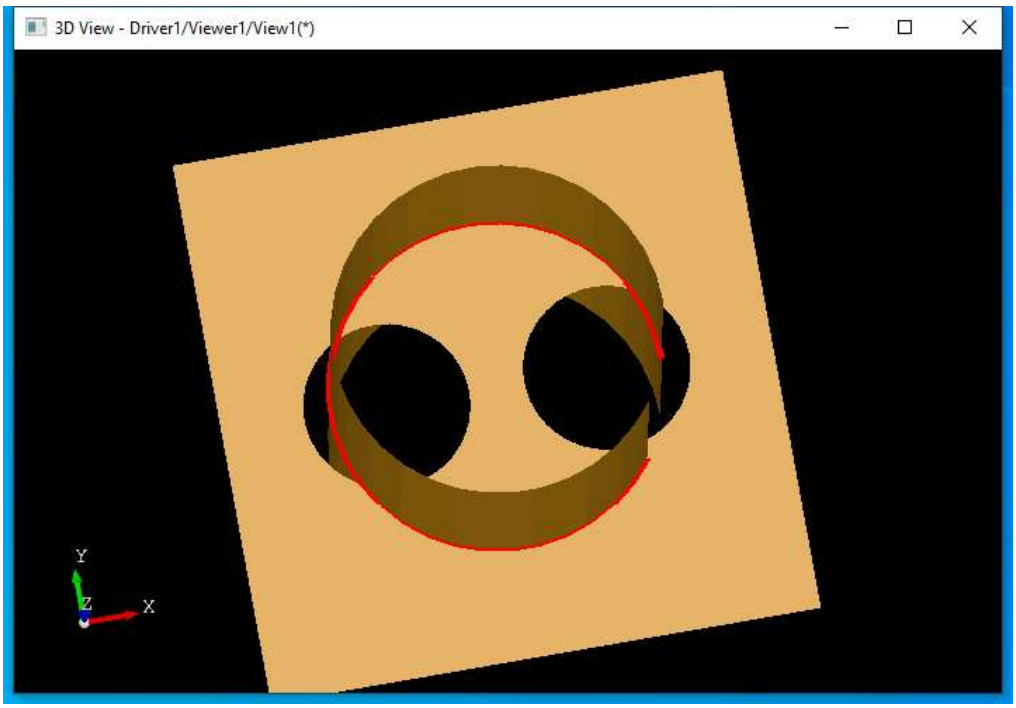
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--IceDream
3. re: OpenCASCADE parameter surface area
@七星重剑
these algorithms are ultimately the application of mathematical theory, and it is quite rewarding to be able to connect theory with practice
--eryar
4. re: OpenCASCADE Incremental Mesh
@七星重剑
OpenCASCADE feels like it is currently the only open-source kernel library in the world with relatively complete styling functions. Thanks.
--eryar
5. re: Introduction to PipeCAD
@七星重剑
Yes, and the mainstream design software is foreign.
--eryar
6. re: EasyMesh - A Two-Dimensional Quality Mesh Generator
It's eye-opening, very useful
--Seven Star Epee
7. re: OpenCASCADE parameter surface area
It's really professional, I admire it
--Seven Star Epee
8. re: OpenCASCADE Incremental Mesh
OpenCASCADE looks pretty good. Congratulations on having a baby :)
--Seven Star Epee

```
IntTools_FaceFace.cxx | BOPAlgo_PaveFiller_7.cxx | BOPAlgo_PaveFiller_6.cxx | BOPAlgo_PaveFiller_1.cxx | BOPTools_A
TKBO | (Global Scope) | Perf
2196  //function : PerformPlanes
2197  //purpose  :
2198  //=======================================================================
2199  void PerformPlanes(const Handle(GeomAdaptor_Surface)& theS1,
2200                     const Handle(GeomAdaptor_Surface)& theS2,
2201                     const Standard_Real TolF1,
2202                     const Standard_Real TolF2,
2203                     const Standard_Real TolAng,
2204                     const Standard_Real TolTang,
2205                     const Standard_Boolean theApprox1,
2206                     const Standard_Boolean theApprox2,
2207                     IntTools_SequenceOfCurves& theSeqOfCurve,
2208                     Standard_Boolean& theTangentFaces)
2209  {
2210
2211    gp_Pln aPln1 = theS1->Plane();
2212    gp_Pln aPln2 = theS2->Plane();
2213
2214    IntAna_QuadQuadGeo aPlnInter(aPln1, aPln2, TolAng, TolTang);
2215
2216    if(!aPlnInter.IsDone()) {
2217      theTangentFaces = Standard_False;
2218      return;
2219    }
2220
2221    IntAna_ResultType aResType = aPlnInter.TypeInter();
2222
2223    if(aResType == IntAna_Same) {
2224      theTangentFaces = Standard_True;
2225      return;
2226    }
2227
2228    theTangentFaces = Standard_False;
```

For overlapping planes, setting theTangentFaces to true means that they are overlapping. Here is a question for everyone to think about: ***How to judge the overlap of freeform surfaces?***

4 Hole

When there are holes on the surface, the intersection line should also be treated to eliminate the intersection line in the hole. When using IntTools_FaceFace to calculate the intersection of two faces, you can see that the intersection line is incorrect, no holes are being handled, and no boundaries of the faces are being treated. As shown in the figure below, the red intersection lines are calculated using IntTools_FaceFace:



When using BOPAlgo_PaveFiller to calculate the intersection line and combine it with PaveBlock, the intersection line is displayed as shown in the following figure:

9. re: RvmTranslator for Linux

Reading bloggers' articles broadens my horizons and inspires me a lot 666

--Seven Star Epee

10. re: Introduction to PipeCAD

There is actually a special "pipeline design software", which is deeply inspired

--Seven Star Epee

11. re: RvmTranslator 7.3

Interesting, pay attention to the WeChat public account

--Seven Star Epee

12. re: OpenCASCADE Plane Intersection

The comment content is long, click on the title to view

--Lao Liu next door

13. re: OpenCASCADE 7.4.0 beta release released

@隔壁老刘

Yes, it was released in 10.1.

--eryar

14. re: OpenCASCADE 7.4.0 beta released

The official version of 7.4.0 has been released on October 1

--Lao Liu next door

15. Re: Wuhan Ocapitade Information Technology Co., Ltd

@ccsdu2009

Thank you

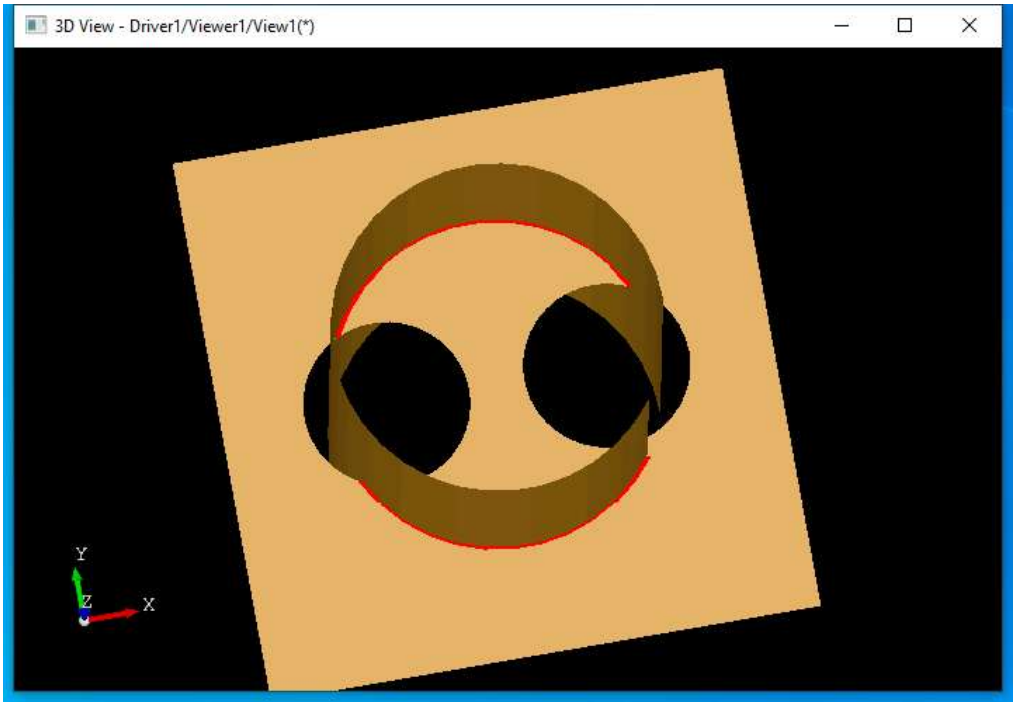
--eryar

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5. B-Spline Basis Functions (16834)

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4. Qt with OpenCascade(25)
5. Boolean Operations in Open Cascade (19)



Although calculating the intersection between two faces is ultimately the IntTools_FaceFace of the call, the correct intersection line requires the use of a class BOPAlgo_PaveFiller. Here are also questions for everyone to think about: ***Why is the intersection range calculated by IntTools_FaceFace incorrect? Why is BOPAlgo_PaveFiller calculate the correct intersection?***

5 Conclusion

To sum up, the results of the intersection of faces in Boolean data may have intersections or intersections. Save the intersection result to FaceInfo. From the simple overlap of two planes, the overlapping state is saved with the variable theTangetFaces. So how to judge the overlap of any two surfaces? Although the intersection of faces provides a IntTools_FaceFace to calculate, the range of intersection lines is not correctly handled, so why can the intersection lines be correctly handled in BOPAlgo_PaveFiller?

I wish you all a happy Mid-Autumn Festival and National Day!

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